

# REFERENSI PERINTAH

## **Linux System Administration**

Praktikum PATI - Mini Server Linux

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# 1. Boot & Service Management

## 1.1 Identifikasi Sistem

| Perintah               | Fungsi                                                    |
|------------------------|-----------------------------------------------------------|
| uname -r               | Menampilkan versi kernel yang sedang berjalan             |
| uname -a               | Info lengkap kernel (hostname, arsitektur, tanggal build) |
| grub-install --version | Menampilkan versi GRUB boot loader                        |
| cat /etc/os-release    | Info distribusi OS (nama, versi, ID)                      |
| hostnamectl            | Info hostname, OS, kernel, arsitektur secara ringkas      |
| lsb_release -a         | Info distribusi Linux (jika tersedia)                     |

## 1.2 Manajemen Service (systemctl)

| Perintah                                 | Fungsi                                      |
|------------------------------------------|---------------------------------------------|
| systemctl status <service>               | Cek status service (active/inactive/failed) |
| systemctl start <service>                | Menjalankan service                         |
| systemctl stop <service>                 | Menghentikan service                        |
| systemctl restart <service>              | Restart service (stop + start)              |
| systemctl reload <service>               | Reload konfigurasi tanpa restart            |
| systemctl enable <service>               | Aktifkan service saat boot                  |
| systemctl disable <service>              | Nonaktifkan service saat boot               |
| systemctl enable --now <svc>             | Enable + langsung start                     |
| systemctl disable --now <svc>            | Disable + langsung stop                     |
| systemctl is-enabled <service>           | Cek apakah service di-enable                |
| systemctl is-active <service>            | Cek apakah service sedang berjalan          |
| systemctl list-units --type=service      | Daftar semua service yang loaded            |
| systemctl list-unit-files --type=service | Daftar semua unit file service              |
| systemctl mask <service>                 | Blokir service agar tidak bisa di-start     |
| systemctl unmask <service>               | Lepas blokir service                        |

## 1.3 Journal & Log

| Perintah                        | Fungsi                                 |
|---------------------------------|----------------------------------------|
| journalctl -u <service>         | Log untuk service tertentu             |
| journalctl -u <service> -f      | Follow log real-time                   |
| journalctl -b                   | Log sejak boot terakhir                |
| journalctl --since '1 hour ago' | Log 1 jam terakhir                     |
| journalctl -p err               | Hanya tampilkan error                  |
| dmesg                           | Kernel ring buffer (log hardware/boot) |
| dmesg   tail -20                | 20 pesan kernel terakhir               |

## 1.4 Custom Service File

Lokasi: `/etc/systemd/system/hello-boot.service`

| Perintah                                 | Fungsi                                  |
|------------------------------------------|-----------------------------------------|
| <code>systemctl daemon-reload</code>     | Reload setelah edit/buat unit file baru |
| <code>systemctl enable hello-boot</code> | Aktifkan custom service saat boot       |
| <code>systemctl start hello-boot</code>  | Jalankan manual                         |
| <code>systemctl status hello-boot</code> | Cek status custom service               |

## 2. Manajemen Disk & Partisi

### 2.1 Informasi Disk

| Perintah     | Fungsi                                                 |
|--------------|--------------------------------------------------------|
| lsblk        | Tampilkan struktur block device (disk, partisi, mount) |
| lsblk -f     | Tampilkan + filesystem & UUID                          |
| fdisk -l     | Detail partisi semua disk                              |
| blkid        | UUID dan tipe filesystem tiap partisi                  |
| df -h        | Penggunaan disk per mount point (human-readable)       |
| df -i        | Penggunaan inode per mount point                       |
| du -sh /path | Ukuran total sebuah direktori                          |
| du -sh /*    | Ukuran tiap direktori di root                          |

### 2.2 Partisi & Format

| Perintah                                          | Fungsi                         |
|---------------------------------------------------|--------------------------------|
| sudo fdisk /dev/sdb                               | Partisi disk sdb (interactive) |
| sudo parted /dev/sdb mklabel gpt                  | Buat GPT label pada disk       |
| sudo parted /dev/sdb mkpart primary ext4 40% 100% | Buat partisi penuh             |
| sudo mkfs.ext4 /dev/sdb1                          | Format partisi sebagai ext4    |
| sudo mkfs.xfs /dev/sdb1                           | Format partisi sebagai XFS     |

### 2.3 Mount & fstab

| Perintah                   | Fungsi                           |
|----------------------------|----------------------------------|
| sudo mount /dev/sdb1 /data | Mount partisi ke /data           |
| sudo umount /data          | Unmount partisi                  |
| mount   grep sdb           | Cek status mount disk sdb        |
| cat /etc/fstab             | Lihat konfigurasi mount permanen |
| sudo nano /etc/fstab       | Edit fstab untuk auto-mount      |
| sudo mount -a              | Mount ulang semua entry di fstab |

### 3. Keamanan Data (Backup & Recovery)

#### 3.1 rsync (Backup)

| Perintah                                          | Fungsi                                         |
|---------------------------------------------------|------------------------------------------------|
| <code>rsync -a /source/ /dest/</code>             | Backup arsip (permissions, timestamps)         |
| <code>rsync -av /source/ /dest/</code>            | Backup verbose (tampilkan file)                |
| <code>rsync -a --delete /src/ /dst/</code>        | Sinkronisasi mirror (hapus file extra di dest) |
| <code>rsync -a /data/ /backup/</code>             | Backup /data ke /backup                        |
| <code>rsync -a /backup/share/ /data/share/</code> | Restore folder share dari backup               |

#### 3.2 Cron (Penjadwalan)

| Perintah                             | Fungsi                          |
|--------------------------------------|---------------------------------|
| <code>crontab -e</code>              | Edit cron untuk user saat ini   |
| <code>sudo crontab -e</code>         | Edit cron untuk root            |
| <code>crontab -l</code>              | Lihat daftar cron user saat ini |
| <code>sudo crontab -l -u root</code> | Lihat cron root                 |
| <code>sudo crontab -r</code>         | Hapus semua cron root           |

**Format Cron:** menit jam hari bulan hari-minggu perintah

| Perintah                  | Fungsi                           |
|---------------------------|----------------------------------|
| <code>0 2 * * *</code>    | Setiap hari jam 02:00            |
| <code>* /5 * * * *</code> | Setiap 5 menit                   |
| <code>0 0 * * 0</code>    | Setiap Minggu jam 00:00          |
| <code>0 3 1 * *</code>    | Tanggal 1 setiap bulan jam 03:00 |

#### 3.3 Simulasi Kegagalan & Recovery

| Perintah                                               | Fungsi                                |
|--------------------------------------------------------|---------------------------------------|
| <code>sudo rm /data/share/Hi.txt</code>                | Simulasi kehilangan file              |
| <code>sudo rsync -a /backup/share/ /data/share/</code> | Restore dari backup                   |
| <code>cat /data/share/Hi.txt</code>                    | Verifikasi file berhasil dikembalikan |
| <code>ls -l /data/share/</code>                        | Cek isi direktori setelah restore     |

## 4. Monitoring Sistem

### 4.1 Monitoring Interaktif

| Perintah            | Fungsi                                           |
|---------------------|--------------------------------------------------|
| htop                | Monitor proses interaktif (CPU, RAM, per-proses) |
| top                 | Monitor proses (bawaan, lebih sederhana)         |
| top -bn1   head -20 | Snapshot non-interaktif top                      |

### 4.2 Monitoring Resource

| Perintah   | Fungsi                                    |
|------------|-------------------------------------------|
| free -h    | Penggunaan RAM dan swap                   |
| vmstat 1 5 | Statistik VM: CPU, memory, IO (5 iterasi) |
| iostat     | Statistik I/O disk dan CPU                |
| sar -u 1 5 | CPU usage per detik (5 kali)              |
| sar -r 1 5 | Memory usage per detik                    |
| uptime     | Load average dan uptime                   |
| nproc      | Jumlah CPU core                           |
| lscpu      | Info detail CPU                           |

### 4.3 Stress Testing

| Perintah                                   | Fungsi                         |
|--------------------------------------------|--------------------------------|
| stress --cpu 2 --timeout 30                | Bebankan 2 CPU selama 30 detik |
| stress --cpu 4 --timeout 60                | Bebankan 4 CPU selama 60 detik |
| stress --vm 1 --vm-bytes 256M --timeout 30 | Stress test RAM 256MB          |
| stress --io 2 --timeout 30                 | Stress test I/O 2 worker       |

## 5. Update Sistem & Keamanan

### 5.1 Update Paket (APT)

| Perintah                                    | Fungsi                                 |
|---------------------------------------------|----------------------------------------|
| <code>sudo apt update</code>                | Refresh daftar paket dari repository   |
| <code>sudo apt upgrade -y</code>            | Upgrade semua paket yang tersedia      |
| <code>sudo apt full-upgrade -y</code>       | Upgrade + hapus paket lama jika perlu  |
| <code>sudo apt autoremove -y</code>         | Hapus paket yang tidak lagi diperlukan |
| <code>sudo apt install &lt;paket&gt;</code> | Install paket baru                     |
| <code>sudo apt remove &lt;paket&gt;</code>  | Hapus paket                            |
| <code>apt list --installed</code>           | Daftar paket yang terinstall           |
| <code>apt list --upgradable</code>          | Daftar paket yang bisa di-upgrade      |
| <code>dpkg -l   grep &lt;keyword&gt;</code> | Cari paket terinstall by keyword       |

### 5.2 Firewall (UFW)

| Perintah                                  | Fungsi                                |
|-------------------------------------------|---------------------------------------|
| <code>sudo ufw enable</code>              | Aktifkan firewall                     |
| <code>sudo ufw disable</code>             | Nonaktifkan firewall                  |
| <code>sudo ufw status</code>              | Status dan rules aktif                |
| <code>sudo ufw status verbose</code>      | Status detail dengan default policies |
| <code>sudo ufw allow 22/tcp</code>        | Izinkan SSH                           |
| <code>sudo ufw allow Samba</code>         | Izinkan Samba (profil aplikasi)       |
| <code>sudo ufw deny 23/tcp</code>         | Blokir Telnet                         |
| <code>sudo ufw delete allow 22/tcp</code> | Hapus rule allow SSH                  |
| <code>sudo ufw reset</code>               | Reset semua rules ke default          |
| <code>sudo ufw app list</code>            | Daftar profil aplikasi yang tersedia  |

### 5.3 Timezone

| Perintah                                                | Fungsi                               |
|---------------------------------------------------------|--------------------------------------|
| <code>timedatectl</code>                                | Status waktu dan timezone saat ini   |
| <code>timedatectl list-timezones</code>                 | Daftar semua timezone                |
| <code>sudo timedatectl set-timezone Asia/Jakarta</code> | Set timezone ke WIB                  |
| <code>date</code>                                       | Tampilkan tanggal dan waktu saat ini |

## 6. Samba File Sharing

### 6.1 Instalasi & Service

| Perintah                               | Fungsi                        |
|----------------------------------------|-------------------------------|
| <code>sudo apt install samba -y</code> | Install Samba                 |
| <code>systemctl status smbd</code>     | Cek status Samba daemon       |
| <code>systemctl restart smbd</code>    | Restart Samba                 |
| <code>testparm</code>                  | Validasi konfigurasi smb.conf |

### 6.2 User & Password

| Perintah                                 | Fungsi                         |
|------------------------------------------|--------------------------------|
| <code>sudo smbpasswd -a adminuser</code> | Tambah/set password Samba user |
| <code>sudo smbpasswd -x adminuser</code> | Hapus Samba user               |
| <code>sudo pdbedit -L</code>             | Daftar Samba users             |

### 6.3 Konfigurasi Share

File: `/etc/samba/smb.conf`

| Perintah                                   | Fungsi                        |
|--------------------------------------------|-------------------------------|
| <code>sudo nano /etc/samba/smb.conf</code> | Edit konfigurasi Samba        |
| <code>testparm -s</code>                   | Validasi dan tampilkan config |

### 6.4 Akses dari Windows

| Perintah                                                | Fungsi                                |
|---------------------------------------------------------|---------------------------------------|
| <code>\\192.168.56.101</code>                           | Akses via File Explorer (address bar) |
| <code>net use Z: \\192.168.56.101\PraktikumShare</code> | Map network drive via CMD             |
| <code>net use Z: /delete</code>                         | Hapus mapped drive                    |

### 6.5 Hak Akses File

| Perintah                                           | Fungsi                      |
|----------------------------------------------------|-----------------------------|
| <code>chmod 755 /data/share</code>                 | Set permissions (rwxr-xr-x) |
| <code>chown adminuser:adminuser /data/share</code> | Set ownership               |
| <code>ls -la /data/share/</code>                   | Lihat permissions detail    |